Summary in Graph

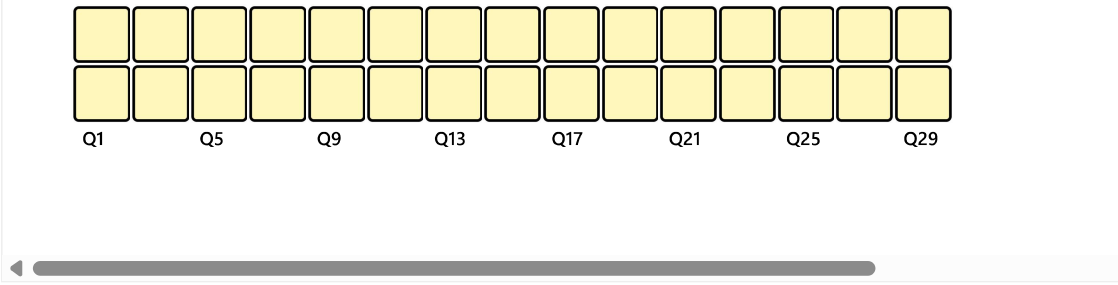
Exam Summary (GO Classes DA Test Series 2026 | Calculus and Optimisation | Subject Wise Test 2)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**30** (10 + 20)
Total Marks:	**50** (10 + 40)
Exam Duration:	**90 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (26) out of 255 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The intervals for which the function $F(x) = x^4 - 4x^3 + 4x^2 + 6$ increases are

A. $x < 0, 1 < x < 2$

B. only $x > 2$

C. $0 < x < 1, x > 2$

D. only $0 < x < 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 04sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let them x, y be positive real numbers such that $xy = 1$. What is the minimum value$x + y$?

A. $\sqrt{2}$
B. 1
C. 2
D. The minimum value does not exist

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f, g : [a, b] \to \mathbb{R}$ be continuous functions. Define

$$h(x) = \min\{f(x), g(x)\}$$

Which of the following statements is **true**?

A. h is not necessarily continuous on $[a, b]$

B. h is continuous only if $f(x) \leq g(x)$ for all $x \in [a, b]$

C. h is continuous on $[a, b]$

D. h is continuous only at points where $f(x) = g(x)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let

$$f(x) = \begin{cases} -x + c & \text{if } x \leq 1 \\ 6 - 2x^2 & \text{if } x > 1 \end{cases}$$

Find the value of c such that $f(x)$ is continuous at $x = 1$.

A. 2
B. 4
C. 5
D. 6

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Your friend has two functions, $f(x)$ and $g(x)$. He calls their product $h(x)$; i.e., $h(x) = f(x)g(x)$. One day he asks you to compute $g'(5)$, but only tells you

- $g(5) = 1$,
- $f(5) = 2$,
- $f'(5) = 3$,
- $h'(5) = 4$.

What is $g'(5)$?

Your Answer: | Correct Answer: 0.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Evaluate the limit:

$$\lim_{x \to 3} \frac{x^2 - x - 6}{x^2 - 9}$$

A. $\frac{5}{6}$
B. 0
C. ∞
D. $\frac{1}{2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

What is the derivative of $g(x) = \frac{1}{f(\sqrt{x})}$?

A. $\frac{f'(\sqrt{x})}{\sqrt{x} f(\sqrt{x})^2}$

B. $\frac{-f'(\sqrt{x})}{\sqrt{x} f(\sqrt{x})^2}$

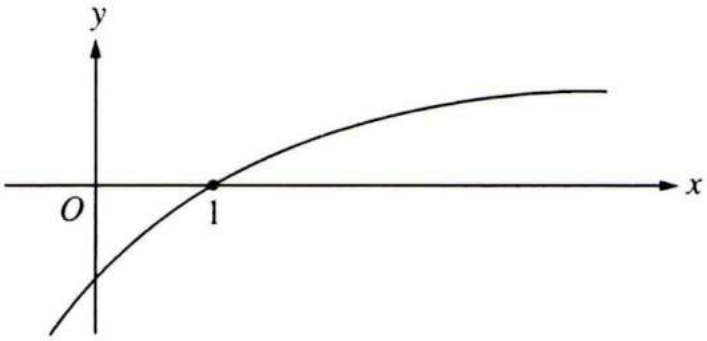
C. $\frac{f'(\sqrt{x})}{2\sqrt{x} f(\sqrt{x})^2}$

D. $\frac{-f'(\sqrt{x})}{2\sqrt{x} f(\sqrt{x})^2}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The graph of a twice-differentiable function f is shown in the figure below. Which of the following is true?



A. $f(1) < f'(1) < f''(1)$

B. $f(1) < f''(1) < f'(1)$

C. $f'(1) < f(1) < f''(1)$

D. $f''(1) < f(1) < f'(1)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

x	0	2	5	9	11
$g(x)$	1	2.8	1.7	1	3.4

The table above shows selected values of a continuous function **g.** For $0 \leq x \leq 11$, what is the fewest possible number of times $g(x) = 2$?

A. One
B. Two
C. Three
D. Four

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

A function $f(x)$ is known to satisfy $f(0) = 1.05$ and $f'(0) = -0.1$. Using the tangent line approximation, what is a reasonable estimate for it $f(0.1)$?

A. 0.95
B. 1.00
C. 1.04
D. 1.10

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let f be a continuous and one-to-one $(1-1)$ function on the interval $[0,1]$, and suppose $f(0) < f(1)$. Which of the following must be true for every $x \in (0,1)$?

A. $f(x) = f(0)$ or $f(x) = f(1)$

B. $f(x) \notin [f(0), f(1)]$

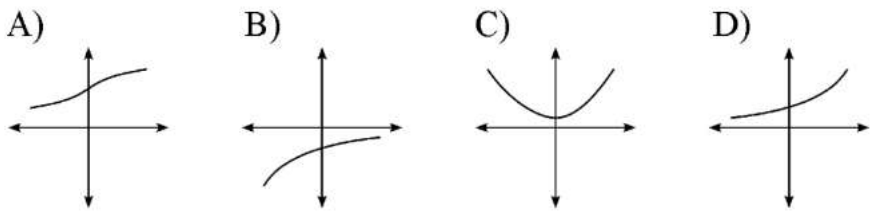
C. $f(0) < f(x) < f(1)$

D. $f(x) > f(1)$ or $f(x) < f(0)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

$f(x), f'(x),$ and $f''(x)$ are all positive for any real number x. Which of the following graphs could be a graph of f ?



Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

What are all values of x for which the function $f(x) = x^3 + 6x^2 + 9x + 1$ is increasing?

A. $(-\infty, -3)\text{only}$

B. $(-3, -1)\text{only}$

C. $(-1, \infty)\text{only}$

D. $(-\infty, -3) \cup (-1, \infty)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If g is a differentiable function such that $g(x) < 0$ for all real numbers x and if $f'(x) = \left(x^2 - 4\right) g(x)$, which of the following is true?

A. f has a relative maximum at $x = -2$ and a relative minimum at $x = 2$.

B. f has a relative minimum at $x = -2$ and a relative maximum at $x = 2$.

C. f has a relative minima at $x = -2$ and at $x = 2$.

D. f has a relative maxima at $x = -2$ and at $x = 2$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Calculus

Evaluate the limit

$$\lim_{x \to \infty} (\sqrt{x+1} - \sqrt{x+2})$$

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $\lim_{x \to a} f(x) = 0$ and $\lim_{x \to a} g(x) = 0$ then

$$\lim_{x \to a} \frac{f(x)}{g(x)}$$

A. 1
B. 0
C. does not exist
D. Not enough information is given to say anything about the limit.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f : \mathbb{R} \to \mathbb{R}$ be a twice continuously differentiable function such that

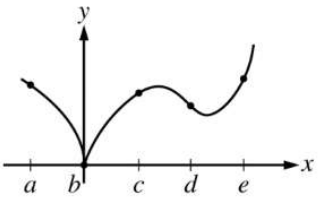
$$f(0) = f(1) = f'(0) = 0.$$

Which of the following statements is necessarily true?

A. $f''(x) = 0$ for all $x \in \mathbb{R}$
B. $f''(0) = 0$
C. There exists some $x \in (0, 1)$ such that $f''(x) = 0$
D. $f''(x) \neq 0$ for any $x \in (0, 1)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



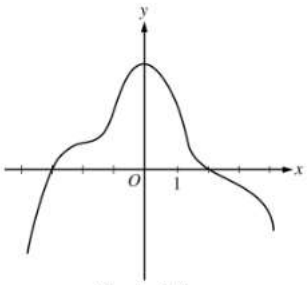
Graph of f

The graph of the function f is shown in the figure above. For which of the following values of x is $f'(x)$ positive and increasing?

A. a
B. b
C. c
D. e

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



Graph of f'

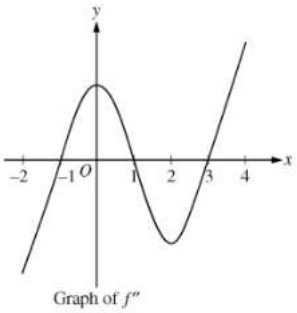
The graph of f', the derivative of the function f, is shown above. Which of the following statements must be true?

I. f has a relative minimum at $x = -3$.
II. The graph of f has a point of inflection at $x = -2$.
III. The graph of f is concave down for 0

A. I only
B. II only
C. I and II only
D. I and III only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



Graph of f''

The graph of f'', the second derivative of f, is shown above for $-2 \leq x \leq 4$. What are all intervals on which the graph of the function f is concave down?

A. $-1 < x < 1$

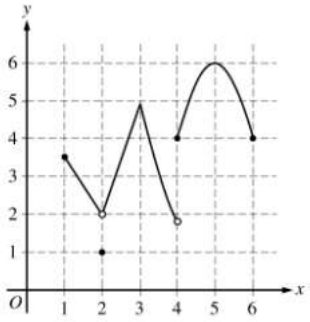
B. $0 < x < 2$

C. $1 < x < 3$ only

D. $-2 < x < -1$ and $1 < x < 3$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



Graph of f

The graph of the function f is shown above. Which of the following statements is false?

A. $\lim_{x \to 2} f(x)$ exists.

B. $\lim_{x \to 3} f(x)$ exists.

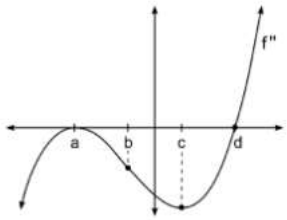
C. $\lim_{x \to 4} f(x)$ exists.

D. The function f is continuous at $x = 3$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Shown is a graph of f'', the second derivative of function f. The curve is given by the equation $f'' = (x-a)^2(x-d)$. The graph of f has inflection points at which values of x ?



A. b only
B. c only
C. a and d
D. d only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The following table shows values of a function $m(x)$ and its derivatives at specific points:

x	0	2
$m(x)$	4	1
$m'(x)$	-1	0
$m''(x)$	0	0
$m^{(3)}(x)$	3	-2
$m^{(4)}(x)$	5	8

What is the degree-4 Taylor polynomial $P_4(x)$ for $m(x)$, centred at $x = 2$?

A. $P_4(x) = 1 - \frac{2}{3!}(x-2)^3 + \frac{8}{4!}(x-2)^4$

B. $P_4(x) = 1 + \frac{3}{3!}(x-2)^3 + \frac{5}{4!}(x-2)^4$

C. $P_4(x) = 1 + \frac{1}{2}(x-2)^2 - \frac{2}{3!}(x-2)^3 + \frac{8}{4!}(x-2)^4$

D. $P_4(x) = 4 - (x-2) + \frac{3}{3!}(x-2)^3 + \frac{5}{4!}(x-2)^4$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Suppose $f(x) = 3x + 2$. The function f is said to be **Lipschitz continuous** if

$$|f(x) - f(y)| \leq L|x - y| \quad \text{for all } x, y$$

What is the **smallest possible Lipschitz constant** L for f ?

A. 2

B. 3

C. 5

D. 0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The function f is continuous on the closed interval $[0, 2]$ and has values that are given in the table below. The equation $f(x) = \frac{1}{2}$ must have at least two solutions in the interval $[0, 2]$ if $k =$

x	0	1	2
$f(x)$	1	k	2

A. 0

B. $\frac{1}{2}$

C. 1

D. 2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Let $f(x, y) = x^2 + y^2$. What is the **maximum rate of change** of f at the point $(1, 2)$, and in which **direction** does it occur?

A. Maximum rate: $\sqrt{5}$; Direction: $\frac{1}{\sqrt{5}}\begin{bmatrix}1\\2\end{bmatrix}$

B. Maximum rate: $\sqrt{20}$; Direction: $\frac{2}{\sqrt{20}}\begin{bmatrix}1\\2\end{bmatrix}$

C. Maximum rate: $2\sqrt{5}$; Direction: $\frac{1}{\sqrt{5}}\begin{bmatrix}1\\2\end{bmatrix}$

D. Maximum rate: 5; Direction: $\begin{bmatrix}1\\2\end{bmatrix}$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Suppose the second partial derivatives of $f(x, y)$ are continuous near a point (a, b), and the second derivative test gives:

$$D = f_{xx}f_{yy} - \left(f_{xy}\right)^2 > 0, \quad \text{and} \quad f_{yy}(a, b) > 0$$

Which of the following is **true**?

A. The test is inconclusive unless we also know the sign of $f_{xx}(a, b)$

B. This implies (a, b) is a local maximum

C. This implies (a, b) is a local minimum
D. The function has a saddle point at (a, b)

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the function:

$$f(x, y) = \frac{1}{3}x^3 - 3x^2 + \frac{y^2}{4} + xy + 13x - y + 2$$

Which of the following statement(s) about the critical points of $f(x, y)$ is/are correct?

A. $f(x, y)$ has critical points at $(3, -4)$ and $(5, -8)$.
B. The critical point $(3, -4)$ is a local minimum.
C. The critical point $(5, -8)$ is a local maximum.
D. The function has no saddle points.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let

$$f(x, y) = 2 - x^2 - 2y^2$$

be a function defined over the unit circle constraint

$$g(x, y) = x^2 + y^2 - 1 = 0$$

Using the method of Lagrange multipliers, what are the **maximum and minimum values** $f(x, y)$ under the given constraint?

A. Maximum$= 2$, Minimum $= 0$

B. Maximum$= 1$, Minimum $= 0$

C. Maximum$= 1$, Minimum $= -1$

D. Maximum$= 2$, Minimum $= -2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f : [0, 5] \to \mathbb{R}$ be continuous function with a maximum at $x = 2$ then

A. the derivative of f at 2 may not exist
B. the derivative of f at 2 must exist and be nonzero
C. the derivative of f at 2 must exist and be zero
D. the derivative of f at 2 can not exist

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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